

6.2 - Standard Form

The equation of a line can be written in two different forms:

1. Slope y-intercept form: $y = mx + b$

where m is the slope, and b is the y-intercept

2. Standard form: $Ax + By + c = 0$

where A , B , and c are integers and A and B are both not zero.

You can change an equation from one form to the other by rearranging the equation.

Example 1:

Write the equation of the line $2x - 3y - 6 = 0$ in slope y-intercept form by isolating the y.

$$2x - 3y - 6 = 0$$

$$\frac{-3y}{-3} = \frac{-2x + 6}{-3}$$

$$y = \frac{-2}{-3}x + \frac{6}{-3}$$

$$y = \frac{2}{3}x - 2$$

Example 2: Write each equation in slope y-intercept form and state the slope and the y-intercept.

a) $3x + 5y - 15 = 0$

$$\frac{5y}{5} = \frac{-3x + 15}{5}$$

$$y = \frac{-3}{5}x + \frac{15}{5}$$

$$y = \frac{-3}{5}x + 3$$

Slope = $m = \frac{-3}{5}$
y-intercept = $b = 3$

b) $7x - 3y + 21 = 0$

$$\frac{-3y}{-3} = \frac{-7x - 21}{-3}$$

$$y = \frac{-7}{-3}x - \frac{21}{-3}$$

$$y = \frac{7}{3}x + 7$$

$$\begin{aligned} \text{Slope} = m &= \frac{7}{3} \\ \text{y-intercept} = b &= 7 \end{aligned}$$

Example 3: Barney's Banquet Facility charges according to the equation $2x - y + 200 = 0$ where x is the number of people attending and y is the total cost.



a) Write the equation in slope y-intercept form.

$$2x - y + 200 = 0$$

$$2x + 200 = y$$

$$y = 2x + 200$$

b) What is the fixed cost?

$$\text{fixed cost} = b = 200$$

$$\boxed{\$200}$$

c) What is the rate of change of the cost?

$$\text{rate of change} = m = 2$$

$$\boxed{\$2 \text{ per person}}$$

d) What is the total cost if 125 people attend a banquet at Barney's?

$$y = 2(125) + 200$$

$$y = 250 + 200$$

$$y = 450$$

$$\boxed{\$450}$$

e) If the total cost is \$920, how many people attend the banquet?

$$920 = 2x + 200$$

$$920 - 200 = 2x$$

$$\frac{720}{2} = \frac{2x}{2}$$

$$\frac{720}{2} = x$$

$$x = 360$$

$$\boxed{360 \text{ people}}$$

1. There are two forms in which the equation of a line can be written. What are they?

Slope y-intercept form: $y = mx + b$

standard form: $Ax + By + C = 0$

2. It is possible to convert an equation from one form to the other by Rearranging the equation.

3. Write the slope-intercept form of the equation of each line:

a) $3x - 2y = -16$

$$-2y = -3x - 16$$

$$y = \frac{-3}{-2}x - \frac{16}{-2}$$

$$y = \frac{3}{2}x + 8$$

c) $9x - 7y = -7$

$$-7y = -9x - 7$$

$$y = \frac{-9}{-7}x - \frac{7}{-7}$$

$$y = \frac{9}{7}x + 1$$

e) $6x + 5y = -15$

$$5y = -6x - 15$$

$$y = \frac{-6}{5}x - \frac{15}{5}$$

$$y = -\frac{6}{5}x - 3$$

g) $11x - 4y = 32$

$$-4y = -11x + 32$$

$$y = \frac{-11}{-4}x + \frac{32}{-4}$$

$$y = \frac{11}{4}x - 8$$

b) $13x - 11y = -12$

$$-11y = -13x - 12$$

$$y = \frac{-13}{-11}x - \frac{12}{-11}$$

$$y = \frac{13}{11}x + \frac{12}{11}$$

d) $x - 3y = 6$

$$-3y = -x + 6$$

$$y = \frac{-1}{-3}x + \frac{6}{-3}$$

$$y = \frac{1}{3}x - 2$$

f) $4x - y = 1$

$$4x - 1 = y$$

$$y = 4x - 1$$

h) $11x - 8y = -48$

$$-8y = -11x - 48$$

$$y = \frac{-11}{-8}x - \frac{48}{-8}$$

$$y = \frac{11}{8}x + 6$$